Acid - Base Indicators

Most dyes that work as acid—base indicators are also weak acids. Therefore let's represent an indicator with the formula HIn. In its un-ionized state, HIn has one color. Its conjugate base, In^- ,has a different color, the more strikingly different the better. In solution, the indicator is involved in a typical acid-base equilibrium:

$$\underset{\substack{\text{Acid form}\\ \text{(one color)}}}{HIn(aq)} \rightleftharpoons H^+(aq) + \underset{\substack{\text{Base form}\\ \text{(another color)}}}{In^-(aq)}$$

The corresponding acid ionization constant, K_{In} , is given by

$$K_{In} = \frac{[H^+][In^-]}{[HIn]}$$

In a strongly acidic solution, when the W concentration is high, this equilibrium is shifted to the left and most of the indicator exists in its "acid form" Under these conditions, the color we observe is that of HIn. If the solution is made basic, the H^+ concentration drops and the equilibrium shifts to the toward In^-, and the color we would observe is that of the "base form' of the indicator.

The observed change in color for an indicator actually occurs over a range of pH values. This is because of the human eye's ability to discern color changes.

As discussed further in Special Topic.1, sometimes a change of as much as:

Suppose that the color of HIn is red and the color of its anion, In^-, is yellow. In terms of the eye's abilities, the ratio of [HIn]/ [In^-] has to be roughly at least as large as 10 to 1 for the eye to see the color as definitely red. And this ratio has to be at least as small as 1 to 10 for the eye to see the color as definitely yellow. Therefore, the color transition that the eye actually sees at the end point occurs as the ratio of [HIn] to [In^-] changes from 10/1 to 1/10.

CamScanner الممسوحة ضوئيا بـ

We can calculate the relationship of pH to pK_a for the indicator at the start of this range as follows. We know that

$$K_{In} = \frac{[H^+][In^-]}{[HIn]}$$

Therefore, at the start of the range, we must have:

$$K_{In} = [H^+] \times \frac{10}{1}$$

$$pK_{In} = pH - \log 10 \qquad pH = pK_{In} - 1$$

At the other end of the range, when $[HIn]/[In^-]$ is 1/10, we have

$$K_{In} = [H^+] \times \frac{1}{10}$$

$$pK_{In} = pH - \log(10^{-1}) \qquad pH = pK_{In} + 1$$

Therefore, the range of pH over which a definite change in color occurs is:

$$pH = pK_{In} \pm 1$$

A range of ± 1 pH unit on either side of the value of pK_{In}, corresponds to a span of 2pH units over which the eye can definitely notice the color change.

How Acid -Base Indicators Work

In a typical acid—base titration, you've seen that as we pass the equivalence point, there is a sudden and large change in the pH. For example, in the titration of HCl with NaOH described earlier, the pH just one-half drop before the equivalence point (when 24.97 mL of the base has been added) is 3.92. Just one-half drop later (when 25.03 mL of base has been added) we have passed the equivalence point and the pH has risen to 10.08. This large swing in pH (from 3.92 to 10.08) causes a sudden shift in the position of equilibrium for the indicator, and we go from a condition where the indicator is in its acid form to a condition in which most is in the base form. This is observed visually as a change in color from that of the acid form to that of the base form.

CamScanner الممسوحة ضوئيا بـ

Selecting the Best Indicator for an Acid—Base Titration

The explanation in the preceding paragraph assumes that the midpoint of the color change of an indicator corresponds to the pH at the equivalence point. At this end point we expect that there will be equal amounts of both forms of the indicator, which means that $[In^-] = [HIn]$. If this condition is true, then

$$K_{In} = \frac{[H^+]\cancel{[In^-]}}{\cancel{[HIn]}}$$

So $$K_{In} = [H^+]$$

Taking the negative logarithms of both sides gives us

$$-\log K_{In} = -\log[H^+] \quad \text{or}$$

$$pK_{In} = pH_{\text{(at the equivalence point)}}$$

This important result tells us that once we know the pH that the solution should have at the equivalence point of a titration, we also know the pK_{In} that the indicator should have to function most effectively. Thus, we want an later whose pK_{In} equals (or is as close as possible to) the pH at the equivalence point. Phenolphthalein, for example, changes a solution from colorless pink as the solution's pH changes over a range of 8.2 to 10.0 (Phenolphthalein is therefore the perfect indicator for the titration of a weak acid by strong base, where the pH of the equivalence point is on the basic side.

CamScanner الممسوحة ضوئيا بـ

Equilibria Involving Polyprotic Acids

Let's begin with the weak diprotic acid, H_2CO_3 . In water, the acid ionizes in two steps each of which is an equilibrium that transfers an H^+ ion to a water molecule.

$$H_2CO_3 = H^+ + HCO_3^-$$

$$HCO_3^- = H^+ + CO_3^{-2}$$

Each step has its own ionization constant, K_a , which we identify as K_{a1} for the first step and K_{a2} for the second. For carbonic acid,

$$K_{a_1} = \frac{[H^+][HCO_3^-]}{[H_2CO_3]} = 4.5 \times 10^{-7}$$

$$K_{a_2} = \frac{[H^+][CO_3^{2-}]}{[HCO_3^-]} = 4.7 \times 10^{-11}$$

Because K_{a1} so much larger than K_{a2} , virtually all the H^+ in the solution comes from the first step in the ionization. In other words,

$$[H^+]_{total} = [H^+]_{first\ step} + [H^+]_{second\ step}$$

$$[H^+]_{first\ step} \gg [H^+]_{second\ step}$$

Therefore, we make the approximation that

$$[H^+]_{total} \approx [H^+]_{first\ step}$$

الممسوحة ضوئيا بـ CamScanner

Ex. 1

Calculate the concentrations of all the species produced in the ionization of 0.05M H_2CO_3 and the pH of the solution.

ANALYSIS

As in any equilibrium problem, we begin with the chemical equations md the K_a expressions.

$$H_2CO_3 \rightleftharpoons H^+ + HCO_3^- \qquad K_{a_1} = \frac{[H^+][HCO_3^-]}{[H_2CO_3]} = 4.5 \times 10^{-7}$$

$$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-} \qquad K_{a_2} = \frac{[H^+][CO_3^{2-}]}{[HCO_3^-]} = 4.7 \times 10^{-11}$$

We will want to calculate the concentrations of H^+, HCO_3^-, and CO_3^{-2} and the concentration of H_2CO_3 that remains at equilibrium.

To obtain $[CO_3^{-2}]$, we will have to use the second step in the ionization. Once again, the large difference between K_{a1} and K_{a2} permits some simplifications. The first involves $[H^+]$, which we have already examined.

$$[H^+]_{total} \approx [H^+]_{first\ step}$$

The second involves $[HCO_3^-]$, for which we can write

$$[HCO_3^-]_{total} = [HCO_3^-]_{\substack{formed\ in \\ first\ step}} - [HCO_3^-]_{\substack{lost\ in \\ second\ step}}$$

However, because $K_{a1} >> K_{a2}$, we expect that

$$[HCO_3^-]_{\substack{formed\ in \\ first\ step}} \gg [HCO_3^-]_{\substack{lost\ in \\ second\ step}}$$

Therefore, we can make the approximation that

$$[HCO_3^-]_{total} \approx [HCO_3^-]_{\substack{formed\ in \\ first\ step}}$$

CamScanner الممسوحة ضوئيا بـ

Solution

Treating H_2CO_3 as though it were a monoprotic acid yields a problem similar to many others we worked. We know some of the H_2CO_3 ionizes; let's call this amount x. If x moles per liter of H_2CO_3 ionizes, then x moles per liter of H^+ and HCO_3^- are formed, and the concentration of H_2CO_3 is reduced by x.

$$[H^+] = [HCO_3^-] = x \qquad [H_2CO_3] = 0.05 - x$$

Substituting into the expression for K_{a1} gives

$$\frac{(x)\,(x)}{(0.050 - x)} = 4.7 \times 10^{-7}$$

Because K_{a1} is so small, we expect that $0.050 - x \approx 0.050$. (This is the approximation we made in solving problems of this kind). The algebra then reduces to

$$\frac{x^2}{0.050} = 4.5 \times 10^{-7}$$

Solving for x gives $x = 1.5 \times 10^{-4}$, so $[H^+]=[HCO_3^-]=1.5 \times 10^{-4}$M. From this we can now calculate the pH.

$$pH = \log(1.5 \times 10^{-4}) = 3.82$$

Now let's calculate the concentration of CO_3^{-2}. We obtain this from the expression for K_{a2}

$$K_{a_2} = \frac{[H^+]\,[CO_3^{2-}]}{[HCO_3^-]} = 4.7 \times 10^{-11}$$

In our analysis, we concluded that we can use the following approximations (where "eq" stands for equilibrium):

$$[H^+]_{eq} \approx [H^+]_{first\,step}$$

$$[HCO_3^-]_{eq} \approx [HCO_3^-]_{first\,step}$$

CamScanner الممسوحة ضوئيا بـ

Substituting the values obtained above gives us

$$\frac{1.5 \times 10^{-4}\,[CO_3^{2-}]}{1.5 \times 10^{-4}} = 4.7 \times 10^{-11}$$

$$[CO_3^{2-}] = 4.7 \times 10^{-11} = K_{a_2}$$

Let's summarize the results. At equilibrium, we have

$$[H_2CO_3] = 0.050\ M$$

$$[H^+] = [HCO_3^-] = 1.5 \times 10^{-4}\ M \text{ (and pH = 3.82)}$$

$$[CO_3^{2-}] = 4.7 \times 10^{-11}\ M$$

Solutions of Weak Diprotic Acids and a Strong Acid

Consider the equilibria for the ionization of oxalic acid.

$$H_2C_2O_4 \rightleftharpoons H^+ + HC_2O_4^-$$

$$HC_2O_4^- \rightleftharpoons H^+ + C_2O_4^{2-}$$

If we increase the H^+ concentration by adding a strong acid, we will shift both of these equilibria to the left, which will lower the $C_2O_4^{-2}$ concentration. If we raise the pH and thereby decrease the H^+ concentration, both equilibria will shift to the right and the concentration of $C_2O_4^{-2}$ will increase. Thus, by controlling the H^+ concentration in the solution, we are able to adjust the oxalate ion concentration.

A simple relationship between the H^+ concentration and the oxalate ion concentration is obtained by multiplying the expressions for K_{a1} and K_{a2}.

$$K_{a_1} \times K_{a_2} = \frac{[H^+]\,[HC_2O_4^-]}{[H_2C_2O_4]} \times \frac{[H^+]\,[C_2O_4^{2-}]}{[HC_2O_4^-]} = \frac{[H^+]^2[C_2O_4^{2-}]}{[H_2C_2O_4]}$$

CamScanner الممسوحة ضوئيا بـ

EX.2

What must the pH of a solution of 0.1 M $H_2C_2O_4$ be to make the $C_2O_4^{-2}$ concenration be 1 X 10^{-4} M?

Solution

We will use the expression for K_{a1} x K_{a2} obtained in the preceding pararaph, using $[H_2C_2O_4]$ = 0.10 M and $[C_2O_4^{-2}]$ 1.0 X 10^{-4} M. The values of K_{a1} and K_{a2} for $H_2C_2O_4$ are given in the table as K_{a1} 6.5 X 10^{-2} and K_{a2} 6.1 X 10^{-5}. Their product is 4.0 X 10^{-6}. Therefore, substituting

$$4.0 \times 10^{-6} = \frac{[H^+]^2(1.0 \times 10^{-4})}{(0.10)}$$

Solving for $[H^+]$ gives

$$[H^+] = 0.063\ M$$
$$pH = -\log(0.063)$$
$$= 1.20$$

To obtain this pH we would probably add a strong acid to the solution.

Practice Exercise

What concentration of $C_2O_4^{-2}$ would exist in 0.2 M $H_2C_2O_4$ if the solution is buffered to a pH of 2 ?

الممسوحة ضوئيا بـ CamScanner

The pH of Solutions of Salts of Polyprotic Acids

We want to deal with salts in which just the anion affects the pH of a solution. Sodium carbonate, Na_2CO_3 , is a typical example.

$$CO_3^{2-}(aq) + H_2O \rightleftharpoons HCO_3^-(aq) + OH^-(aq) \qquad (1)$$

The bicarbonate ion is the other product of the equilibrium, however, and HCO_3^- is also a Bronsted base. So it generates more OH^- ion by the follow equilibrium.

$$HCO_3^-(aq) + H_2O \rightleftharpoons H_2CO_3(aq) + OH^-(aq) \qquad (2)$$

How should we calculate the pH of a solution which OH^- is produced by more than one equilibrium like this?

$$K_{b_1} = \frac{[HCO_3^-][OH^-]}{[CO_3^{2-}]} = 2.1 \times 10^{-4}$$

This K_{b1} value for CO_3^{2}, 2.1 X 10^{4}, is calculated from the relationship applies to any conjugate acid — base pair, $K_aK_b = K_w$, or

$$K_b = \frac{K_w}{K_a}$$

The conjugate acid of CO_3^{-2} is HCO_3^-, and its acid ionization constant is the K_{a2} for carbonic acid: $K_{a2} = 4.7 \times 10^{-11}$ (from table). Therefore,

$$K_{b_1} = \frac{K_w}{K_{a_2}} = \frac{1.0 \times 10^{-14}}{4.7 \times 10^{-11}} = 2.1 \times 10^{-4}$$

For the base ionization constant of HCO_3^-, we use the second equilibrium above, Equation.2, and write

$$K_{b_2} = \frac{[H_2CO_3][OH^-]}{[HCO_3^-]} = 2.2 \times 10^{-8}$$

The value of K_{b2} is also found by using $K_aK_b = K_w$. H_2CO_3 is the conjugate acid of HCO_3^- , so we must use $K_{a1} = 4.5 \times 10^{-7}$ (from table). Therefore, the K_{b2} value is

CamScanner الممسوحة ضوئيا بـ

calculated as follows.

$$K_{b_2} = \frac{K_w}{K_{a_1}} = \frac{1.0 \times 10^{-14}}{4.5 \times 10^{-7}} = 2.2 \times 10^{-8}$$

Now we can compare the two K_b values. For CO_3^{-2}, K_{b1}=2.1 x10^{-4}, and for the (much) weaker base, HCO_3^-, K_{b2} is 2.2 X 10^{-4}. Thus CO_3^{-2} has a K_b nearly 10,000 times that of HCO_3^-.

$$[OH^-]_{total} = [OH^-]_{\substack{\text{formed in}\\ \text{first step}}} + [OH^-]_{\substack{\text{formed in}\\ \text{second step}}}$$

$$[OH^-]_{\substack{\text{formed in}\\ \text{first step}}} \gg [OH^-]_{\substack{\text{formed in}\\ \text{second step}}}$$

$$[OH^-]_{total} \approx [OH^-]_{\substack{\text{formed in}\\ \text{first step}}}$$

Thus, when we calculate the pH of a solution of a basic anion of a polyprotic acid, we may work exclusively with this anion and not with others that can form in additional equilibria. Let's study an example of how this works.

EX.3

What is the pH of a 0.15 M solution of Na_2CO_3?

Analysis

We can ignore the reaction of HCO_3^- with water so the only relevant equilibrium is:

$$CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^- \qquad K_{b_1} = \frac{[HCO_3^-][OH^-]}{[CO_3^{2-}]} = 2.1 \times 10^{-4}$$

(The value of K_{b1} was calculated in the discussion preceding this example.) Also, the initial concentration of CO_3^{-2} (0.15 M) is larger than 400 x K_{b1} ,so we can use 0.15M as the equilibrium concentration of CO_3^{-2}

CamScanner الممسوحة ضوئيا بـ

Solution

Some of the CO_3^{-2} will react; we'll let this be x mol L^{-1} . The concentration table, then, is as follows.

$$CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$$

Initial concentrations (M)	0.15	0	0
Changes in concentrations caused by the ionization (M)	$-x$	$+x$	$+x$
Final concentrations at equilibrium (M)	$(0.15 - x) \approx 0.15$	x	x

Let's now insert the values in the last row of the table into the equilibrium expression for the carbonate ion.

$$\frac{[HCO_3^-][OH^-]}{[CO_3^{2-}]} = \frac{(x)(x)}{0.15} = 2.1 \times 10^{-4}$$

$$x = 5.6 \times 10^{-3}$$

Notice that the value of x really is sufficiently smaller than 0.15 to justify our using 0.15 as a good approximation of (0.15 - x).Because $x=5.6 \times 10^{-3}$

$$[OH^-] = 5.6 \times 10^{-3} \text{ mol } L^{-1}$$

$$pOH = -\log(5.6 \times 10^{-3}) = 2.25$$

Hence,

$$pH = 14 - 2.25 = 11.75$$

Thus, the pH of 0.15 M Na_2CO_3 is calculated to be 11.75. Once again we see that an aqueous solution of a salt composed of a basic anion and a neutral cation is basic.

CamScanner الممسوحة ضوئيا بـ